The Interview-Compatible Recommender: A Measuring Instrument for Cold-Start Conversational Recommendation

Thesis Chapter A (v2)

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Abstract

Before one can study *how* to interview a cold-start user—which questions to ask, in what order—one needs a recommender that can *be* interviewed: a model that folds an arbitrary-length, mixed-channel set of answers into its belief and re-ranks a large catalogue, without per-user retraining and without trading away full-profile accuracy. We argue that this recommender is a *measuring instrument* for the entire elicitation-policy literature, and that its requirements are more demanding than any single published system meets. We state five requirements—**R1** state-of-the-art full-profile accuracy, **R2** arbitrary-length cold-start fold-in, **R3** channel-agnostic input (items, out-of-catalogue concepts, entities, and continuous directions through one interface), **R4** an uncertainty-native belief, and **R5** monotonicity (a truthful answer never lowers ranking quality)—and an acceptance battery (per-build gates **G0**–**G9** plus one-time certifications **C1**–**C3**) that operationalises them. A survey of six model families shows the literature is *bifurcated*: the models that hit the full-profile bar (shallow linear/autoencoder recommenders such as EASE [21] and RecVAE [20]) are item-only point estimators, while the models with an uncertainty-native, multi-channel belief (soft-attribute Gaussian elicitation [2], unified item+attribute bandits [12], conjugate critique folds [24]) do not report full-profile accuracy at the R1 bar. We are aware of no published system that occupies the conjunction of all five. The instrument we propose to close this gap is a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian belief layer over its latent, into which every channel folds as a linear observation; three of its input legs—signed/graded values, out-of-catalogue concepts, and continuous direction tokens—are architecturally inexpressible in the item-indicator basis of the R1-bar models. This chapter contributes the requirement framework, the taxonomy and gap analysis, the instrument's design, and the experimental protocol (a metric-certification bridge to the published ML-20M numbers, then G0/G1/G2 on the ML-25M harness). Our in-house RecVAE realisation sets the current bar at full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M protocol; the graded-native interview tower is training and all instrument results are marked **[TODO:]** pending certification.

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1 Introduction: the recommender as a measuring instrument

A *cold-start* conversational recommender must elicit taste from a short interview and then rank a large catalogue. Two questions are entangled: *what to ask* (the elicitation policy) and *how to turn answers into recommendations* (the recommender). The elicitation-policy literature is large and active, but almost every comparison in it is confounded: a policy's apparent value depends on the recommender it drives, and reported gains routinely conflate a better *question-selection strategy* with a better *answer-ingestion model*. Before we can measure elicitation strategies in isolation, we need to fix the recommender—and not just any recommender, but one strong and flexible enough that a good interview is not wasted on it. We call such a recommender an *interview-compatible instrument*.

Five requirements. An instrument fit for measuring conversational elicitation must simultaneously satisfy:

R1 — full-profile SOTA. On a full interaction history it must rank at the level of the strongest shallow collaborative-filtering baselines (EASE [21], RecVAE [20]), so that elicitation is measured against a recommender nobody can dismiss as weak [7].

R2 — any-length cold-start fold-in. It must ingest an evidence set of *any* cardinality, from 0 (fully cold) upward, without per-user retraining.

R3 — channel-agnostic input. Items, out-of-catalogue concepts, entities, and free text (and, for the continuous policy, direction tokens) must all fold through *one* interface as arbitrary embeddings, not a fixed categorical schema.

R4 — uncertainty-native belief. It must carry an explicit covariance over the user belief, not just a point estimate, so that a policy can act on *what it does not yet know*.

R5 — monotonicity. A truthful answer must never *lower* ranking quality; adding evidence should, in expectation, only help.

The bifurcation observation. The central empirical fact this chapter documents is that R1 and {R3, R4, R5} are *anti-correlated across the published literature*. The models that hit the full-profile bar—closed-form linear autoencoders (EASE [21], EDLAE [22]), amortised VAEs (Mult-VAE [13], RecVAE [20]), and training-free graph filters (GF-CF [19], BSPM [4], Turbo-CF [17])—are *item-only point estimators* on a one-hot indicator basis. The models with an uncertainty-native, multi-channel belief—soft-attribute Gaussian elicitation [2], item+attribute Thompson bandits [12], conjugate critique folds over a frozen factorisation [24], and per-item Bayesian NL elicitation [1]—do *not* report full-profile accuracy at the R1 bar. We call these the *accuracy corner* and the *belief corner*. No published system we are aware of occupies both (§4).

Contributions. This chapter is the instrument paper of the thesis. It does not claim a new elicitation policy (that is the companion method chapter); it claims the *recommender that makes policy measurement possible*. Concretely:

1. **The conjunction as the contribution.** We frame R1∧R2∧R3∧R4∧R5 as a single requirement no published system meets, and give a six-family taxonomy and per-near-miss gap analysis (§3, §4) that locates exactly which requirement each candidate fails. We say we are *aware of none*, never that there *is* none.
2. **Three architecturally-inexpressible legs.** We isolate the three input channels that are impossible (not merely untuned) in the item-indicator basis of every R1-bar model: (i) signed/graded per-entity values, (ii) out-of-catalogue concepts, and (iii) continuous direction tokens. These are the narrow, defensible novelty legs of the channel-agnostic interface (§4, §5).
3. **The instrument.** A design for a *frozen* certified shallow-SOTA tower with an analytic conjugate-Gaussian precision-accumulator belief layer over its latent, into which every channel folds as a linear observation (§5). The frozen-tower+exact-conjugacy shape is the one mechanism with a principled monotonicity story [24], and it is what separates us from the co-trained, approximate-posterior near-miss of Bıyık et al. [2].
4. **An acceptance battery as a reusable certificate.** We define three headline outcome gates—full-strength preserved (G0), posterior strictly shrinks per question (G1), and cold-start per-question NDCG rises on *full and tail* versus random (G2)—and, because these outcome gates admit several false positives (a sign-blind fold, a circular answer model, a decorative covariance), extend them into a full battery of per-build gates G0–G9 plus one-time certifications C1–C3 that add the semantic, artifact-control, and answer-model-transfer checks. We propose the battery as a portable certificate any interview-compatible recommender should pass (§2, §6).

We are careful throughout to distinguish what is *new* (the conjunction; the frozen-certified-tower wedge; the acceptance battery on a SOTA tower) from what is *pre-empted* and must be cited (attributes-as-directions plus a Bayesian fold [2]; unified item+attribute belief [12]; conjugate folds over a frozen factorisation [24]; permutation-invariant set-encoders with information-gain acquisition [16]; value-carrying set-input cold-start recommenders [14]; the non-degradation-of-information principle, Good 1967 / Blackwell). "Uncertainty-native recommender" alone is a crowded 2024–25 area [3] and is *not* claimed as novel.

2 Requirements and gates

We now state the requirements and gates formally. Let the catalogue have n items and let the frozen tower expose a d-dimensional latent in which each item i has a factor row $\mathbf{Q}_i \in \mathbb{R}^d$. An *evidence set* after t questions is a set of observations $\mathcal{E}_t = \{(\boldsymbol{\phi}_j, y_j)\}_{j=1}^{t}$, where $\boldsymbol{\phi}_j \in \mathbb{R}^d$ is the latent direction of the queried entity (an item factor, a concept centroid, an entity embedding, or a continuous direction token) and $y_j \in \mathbb{R}$ is the (possibly graded, possibly signed) answer. The instrument maintains a belief $b(\mathcal{E}_t) = (\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ over the user latent $\mathbf{u} \in \mathbb{R}^d$ and scores item i by a fixed read-out $s_i(\boldsymbol{\mu}_t)$ of the base tower.

Requirements (formal).

R1. With $\mathcal{E}$ equal to a user's full history, $\mathrm{NDCG@10}\big(s(\boldsymbol{\mu}(\mathcal{E}))\big)$ ties the strongest shallow baseline on a shared harness (Def. 1).

R2. $b(\mathcal{E}_t)$ is defined and computable for every $t \geq 0$ with no gradient step at inference; $t{=}0$ yields the prior belief (a non-personalised ranker).

R3. The update from $\mathcal{E}_t$ to $\mathcal{E}_{t+1}$ depends on the new observation only through the pair $(\boldsymbol{\phi}_{t+1}, y_{t+1}) \in \mathbb{R}^d \times \mathbb{R}$; the operator is identical for every channel, so any entity with an embedding is a legal observation—including entities absent from the catalogue.

R4. $\boldsymbol{\Sigma}_t$ is an explicit, queryable covariance; a policy may read $\boldsymbol{\Sigma}_t$ to choose the next question.

R5. For a truthful answer y_{t+1}, $\mathbb{E}\big[\mathrm{NDCG@10}(s(\boldsymbol{\mu}_{t+1}))\big] \geq \mathrm{NDCG@10}(s(\boldsymbol{\mu}_t))$. We treat R5 as an *empirical* property to be demonstrated per step, not a theorem for a learned ranker (§7).

Definition 1 (Ties SOTA). *A recommender* ties *a baseline on a metric if their scores are within the paired per-user bootstrap confidence interval on one shared harness (same users, items, split, metric depth). A number computed on a different metric/split (e.g. NDCG@10 on ML-25M vs. NDCG@100 on the ML-20M strong-generalisation split) is* not *a tie until bridged (§6.2).*

Gates (acceptance battery). The three headline gates below (G0/G1/G2) are *outcome* gates: they certify that the full-profile strength is preserved, that the posterior is usable, and that the cold-start curve rises. On their own they are insufficient (see below), so the instrument is accepted only against the full battery—per-build gates G0–G9 (every build, all cheap) and three one-time certifications C1–C3. Table 1 is the compact reference; the definitions follow.

Per-build gates.

G0 — full strength & identity. Evaluating on the belief mean with the full profile reproduces the frozen tower's full-profile NDCG@10 (full *and* tail)—a paired-bootstrap tie, or bit-identity under the frozen-fold design—so the belief layer adds no full-profile regression (certifies R1 survives).

G1 — usable posterior. The posterior covariance shrinks in expectation over confident answers, with *refusals explicitly exempt* (a correct model does not shrink on a refusal); the shrinkage is directional (at least twice as large along the queried $\boldsymbol{\phi}_{t+1}$ as elsewhere) and the belief scale is calibrated against held-out NLL (certifies R4).

G2 — cold-start curve rises on full and tail. Starting cold, per-question full and tail NDCG@10 beats a random-question control judged by the *curve/endpoint at budget* (an AUC criterion, *not* every-step significance), and must strictly *increase* from the first to the last turn (not merely fail to drop); plus an out-of-envelope canary that a single true answer from cold helps on *every* channel. This is the operational form of R2$\wedge$R5.

G3 — sign & intensity fidelity. Negating every answer's sign lowers NDCG (CI excluding zero, per channel); neutralising values to "indifferent" while freezing membership costs NDCG; and quality is monotone in stated intensity, so a binarising ablation must lose.

G4 — confidence & refusal. The fitted per-answer precisions order $\alpha(\text{refuse}) < \alpha(\text{vague}) < \alpha(\text{know-well})$, collapsing the confidence levels loses NDCG, and a refusal burns the turn while moving the belief by approximately nothing.

G5 — concept/channel specificity. Folding a concept ranks its member items above matched non-members (member-lift AUC > 0.8) *and* beats that concept's own popularity projection (top principal component partialled out), with a dedicated channel at least matching a trained member-bag injection.

G6 — existential controls. Wrong-user, shuffled, and placebo-constant answers through the identical pipeline gain approximately nothing relative to the true-answer arm, and duplicating an answer changes nothing (the belief responds to information, not cardinality).

G8 — $\mathbf{\Sigma}$ load-bearing & set coherence. Replacing $\mathbf{\Sigma}$ with an isotropic $c\mathbf{I}$ *degrades* the acceptance-deciding number (else the R4 claim is withdrawn), and the belief and ranking are invariant to answer order and to conflicting re-asks.

G9 — answer-ingestion isolation. On a *fixed*, user-independent question set the curve still rises with t from answer content alone, separating instrument *ingestion* from question-selection *policy*.

One-time certifications (per instrument, not per build):

C1 — published-number bridge. The implementations snap to the published full-profile numbers on the canonical split (*partially met*: EASE and RecVAE reproduce to the third decimal on the ML-20M Liang split, §6.2; Mult-VAE/DAE pending).

C2 — answer-model transfer (non-circularity). The G2 gain survives under a *held-out behavioural* answer model plus injected label noise, at $\geq$ a stated fraction of the self-model gain with a CI excluding zero, and a firewall assert guarantees that recommender-geometry answers appear nowhere in training or evaluation. This is the load-bearing certification: without it a G0–G2 pass is consistent with an instrument inverting its own simulator.

C3 — human round-trip. At least one study takes rendered questions $\rightarrow$ graded human answers $\rightarrow$ downstream NDCG. Deferred; carried as a project-wide external-validity obligation (§7).

Why the outcome gates alone are insufficient. G0/G1/G2 are outcome gates, and several broken instruments pass all three. A *sign-blind* fold that updates its belief from only the *direction* of the queried entity but never the answer *value* still reproduces the frozen tower (G0), still shrinks $\mathbf{\Sigma}$ along $\boldsymbol{\phi}$ per question (G1), and still beats a random-question control (G2)—because beating random certifies that a good question was *selected*, not that the answer was *ingested*; only the sign-flip test (G3) and the fixed-question ingestion isolation (G9) expose it. A *circular* answer model that generates answers from the instrument's own latent geometry inflates the G2 curve with no transferable signal; only the held-out behavioural answer model and firewall of C2 close it. And a *decorative* covariance that is maintained but never enters the acceptance-deciding number passes G0–G2 trivially; only replacing $\mathbf{\Sigma}$ with an isotropic matrix and demanding a degradation (G8) shows whether the uncertainty is load-bearing. The battery's semantic and artifact-control gates (G3–G9) and the transfer certification (C2) exist precisely to close these outcome-gate false positives.

We report *both* full and tail NDCG@10 for every gate (a project reporting rule: full-only or tail-only hides the effect, which is typically much larger on the tail than on the full catalogue). All gate results in this chapter are **[TODO:]** pending the certified run of §6.

3 Related work: a six-family taxonomy

We organise interview-relevant recommenders into six families and score each against R1–R5 (Table 2). The reading is consistent across families: no family is strong on both the accuracy axis (R1) and the belief-channel axis (R3–R5). Figure 1 plots the same fact quantitatively, with measured full-profile accuracy on one axis. Table 2 is the compact conceptual overview; the chapter's anchor is the per-system *master table* (Table 3), which classifies every reasonable candidate against R1–R5 *and*, where one exists, prints its full/tail NDCG@10 on our shared ML-25M ruler—so that the bifurcation is not merely asserted from published claims but is visible in the one column of measured numbers.

No dash is an unexplained omission. Every unmeasured row carries one of three dispositions. (Systems with neither a measurable number nor a distinct capability profile—e.g. SLIM, strictly dominated by its closed-form successor EASE in every published comparison—are cited in the text but given no row.) *Run planned*: the graph filters and sequence models have public code and will be measured on this ruler in the next experimental wave. *Best-effort measured / planned*: for systems whose full-profile core is runnable but whose exact recipe is unpublished or benchmark-bound, we report a best-effort replication of that core with every substitution stated. Three are now *measured*, and they land weak: a Gaussian belief over an MF latent—the shared posterior core of Bıyık (no public implementation; their embedding is co-trained, so the substitution, if anything, flatters the original) and of ConTS's Thompson posterior—measures 0.2019/0.1200 full/tail, *below* even iALS; and the node recommender that is the Golbandi tree's core measures 0.3065/0.1895. A trained Partial-VAE for EDDI and MF-seed reconstruction for RBMF/Functional-MF remain best-effort planned. That the belief corner's runnable cores measure this far below the accuracy corner sharpens the bifurcation rather than softening it. *Not runnable by construction*: PEBOL maintains per-item Beta posteriors over an LLM-scored shortlist of ∼100 items and contains no full-catalogue ranker to run; GATE elicits but does not rank; zero-shot LLM ranking of an 18k catalogue is neither computationally nor methodologically meaningful at full-profile scale (and its collaborative weakness is already documented [9]); BCIE requires its authors' knowledge graph, and a port would substitute so much that the number would no longer be theirs—its conjugate-fold core is covered by the Bıyık-style best-effort arm. ICF is the one row whose full-profile core is *already* measured: its underlying model is Bayesian probabilistic MF, so its full-profile ceiling is the measured iALS row (0.2442).

The measured-accuracy column falls as you enter the belief corner. Read Table 3 down its measured-accuracy column. Every *competitive* number—a full/tail NDCG@10 at or near the frontier on our shared ML-25M ruler—sits in block (a) or (b): the item-only linear and amortised recommenders (RecVAE 0.3540, EASE 0.3476, EDLAE 0.3433). The belief corner, block (c), is either *blank* or *measurably weak*. Where its cores are runnable at all they land at the bottom of the column: the shared Gaussian-belief-over-MF core of Bıyık and ConTS at 0.2019/0.1200 (below iALS and kNN), the Golbandi node recommender at 0.3065/0.1895—and every one of these carries an ✗ on R1. The rest of the corner—EDDI, RBMF/Functional-MF, BCIE, UNICORN, PEBOL, GATE, the LLM-zero-shot CRS—has a blank accuracy column entirely. This is not an accident of our harness. Most CRS and elicitation papers never report full-catalogue top-N accuracy at all (they report elicitation RMSE, cold-start MRR/MAP on a shortlist, or critique metrics), and several *cannot produce it by construction*: a per-item Beta posterior (PEBOL) or the decision tree itself (Golbandi, whose node recommender we measure only as a best-effort core) does not induce a ranking over the whole catalogue, and an LLM CRS needs an API call per candidate to score one.

Conversely, every system at the frontier is item-only and point-estimate—✗ on R3 and R4. The two blocks are disjoint: no row combines a frontier accuracy number with ✓s across R3–R5. The top-right cell of the gap map—all five requirements met *and* a competitive full-profile number—is occupied only by the target row, which is a design, not a result. That empty conjunction is the chapter's thesis, and the master table is where it is impossible to miss.

The accuracy corner (families ii–iii). On the canonical ML-20M strong-generalisation split (10k validation / 10k test held-out users, NDCG@100), the full-profile frontier has been *flat since 2019–2020* and is held by shallow models, not deep nets: EASE [21] at NDCG@100 0.420, Mult-VAE [13] at 0.426, RecVAE [20] at 0.442 (top of this split), with EDLAE [22] in the same class; SLIM (0.401) and WMF/iALS (0.386) are the classic floors. Dacrema et al. [7] is the binding warning: most deeper "SOTA" gains vanish on this exact split. The graph-filter leaders (GF-CF [19], BSPM [4], Turbo-CF [17]) are the 2023–24 speed+accuracy front on Gowalla/Yelp/Amazon but do not report this ML-20M split, so they are admissible to our accuracy tier only if re-run on our harness. All of family (ii)/(iii) are item-only: their input is a one-hot (or set-indicator) interaction vector, which *cannot encode* a signed value, an out-of-catalogue concept, or a continuous direction—R3 is an architecture property here, not a tuning gap.

The belief corner (family iv, and vi). Bıyık et al. [2] maintain a multivariate-Gaussian belief over a user embedding and treat both item comparisons and "soft attributes" (e.g. "scarier") as directions (concept-activation vectors) in the item space, selecting queries by expected value of information—the nearest thing to R2+R3+R4 together. ConTS [12] unifies items and attributes as Thompson-sampling arms with a posterior over a cold-start user. BCIE [24] folds critiques into a *frozen* factorisation by exact conjugate-Gaussian updates—the closest published analogue of our belief mechanism. PEBOL [1] keeps a per-item Beta posterior updated by LLM-extracted aspects and NLI scoring, reporting cold-start MRR@10 0.27 versus a 0.17 baseline. None of these reports full-profile accuracy at the EASE/RecVAE bar; each is a belief layer, not a certified tower.

The set-encoder and cold-start-NP precedents (families i, iii). EDDI/Partial-VAE [16] is the permutation-invariant set-encoder over an arbitrary observed subset with information-gain acquisition—exactly R2's architecture, but demonstrated outside recommendation and without a full-profile CF result or an attribute/concept channel. TaNP [14] gives a neural-process, stochastic-process uncertainty over a value-carrying set input for cold-start—the R2+R4 shape, but not at full-profile SOTA nor channel-agnostic. These are cited as the mechanism precedents our design instantiates, not as rivals for the conjunction.

The gap map. Figure 1 places every system on two axes: *interview-compatibility* on the x-axis—a coarse ordinal capability count, the number of R1–R5 marks a system satisfies in the master table (✓=1, $\sim$=0.5)—and *measured full-profile NDCG@10 on our ML-25M ruler* (R1) on the y-axis. The measured points trace a *descending* frontier: accuracy peaks at the item-only accuracy corner (x=2.5) and falls as interview-compatibility rises, so the top-right—high compatibility *and* a full-profile number—is empty.

4 Gap analysis

The verdict. No single published system satisfies all of {R1 SOTA-full-profile $\wedge$ R2 any-length set $\wedge$ R3 channel-agnostic via arbitrary embeddings $\wedge$ R4 uncertainty-native $\wedge$ R5 monotone}.

The binding near-miss is Bıyık et al. [2]: it fails R1 (a co-trained item space, not a frozen certified RecVAE-class tower, and no full-profile benchmark) and gives only an approximate, "generally not Gaussian" posterior; every other candidate fails R3 or R4 outright. **We are aware of none** that occupies the conjunction—and we state it exactly that way, never as "there is none."

Per-near-miss failure analysis.

- **Bıyık et al. 2023** [2] — has R2 + R3 (items and soft attributes as directions) + R4 (Gaussian belief). *Fails R1*: the item space is co-trained with the belief model, not a certified full-profile CF tower, and accuracy is reported on their elicitation task, not the EASE/RecVAE bar. R3 is *attribute directions*, not arbitrary embeddings / open concepts / continuous query tokens. R5 is approximate (sampled EVOI over a non-Gaussian posterior). This is the central threat, and our wedge is exactly (a) frozen certified tower vs. co-trained, (b) exact conjugacy vs. approximate posterior, (c) arbitrary / open / continuous tokens vs. a fixed attribute set.
- **ConTS** [12] — R2 + R4 + a unified item/attribute action space. *Fails R3* (a fixed categorical attribute schema; no arbitrary embedding, no continuous or open token) and *R1* (a bandit over arms, not a full-profile encoder).
- **EDDI / Partial-VAE** [16] — R2 + R4 (information-gain) + set input. *Fails R1* (not a recommendation model / no CF SOTA) and *R3* (no attribute or concept channel as demonstrated).
- **RecVAE / EASE** [20, 21] — *R1 yes. Fail R3* (an item-only one-hot / indicator basis—cannot express a concept, a signed value, or a continuous direction) and *R4* (point estimate; RecVAE's latent posterior is item-only and is not used as an actionable belief).
- **PEBOL** [1] — R4 (per-item Beta) + R3 (LLM aspects) + strong cold-start. *Fails R1* (no full-profile CF ranking) and its R4 is *per item*, not a shared user covariance; no monotonicity guarantee.
- **SASRec / BERT4Rec** [10, 23] — R2 (re-encode the sequence). *Fail R3, R4, R5* (item-only, order-sensitive, point estimate, non-monotone—a later token can reorder everything). Token input is trivially satisfied but is *not* value-carrying or mixed-type, so "interview = tokens" does not hold; and BERT4Rec's fold-in advantage carries a replicability caveat [18].

The three inexpressible legs. Three input channels are *architecturally impossible* (not merely untuned) in the item-indicator basis of every R1-bar model: (i) graded/signed per-entity values (the basis has no sign and no magnitude—a "dislike" is not representable as a negative one-hot without changing the likelihood), (ii) out-of-catalogue concepts (there is no column for an entity absent from the catalogue), and (iii) arbitrary continuous direction tokens. Folding these as *linear observations in a frozen latent* is the one leg of the conjunction we find nowhere in the literature.

Pre-empted—cited, never claimed as first. We must credit, and do not claim priority over: attribute-answers-as-directions plus a Bayesian fold [2]; unified item+attribute belief and value-of-information [2, 12]; conjugate-Gaussian folds over a frozen factorisation [24]; permutation-invariant set-encoders with information-gain acquisition [16]; value-carrying set-input cold-start recommenders [14]; the non-degradation-of-information principle (Good 1967 / Blackwell); and the now-crowded "uncertainty-native recommender" area [3]. Our defensible novelty is only the *conjunction* of all five requirements in one frozen certified tower, plus the three inexpressible legs and the G0–G2 protocol.

5 The instrument

The design follows directly from the gap analysis: use the shallow frozen tower that already holds the accuracy corner (R1, and the flat-since-2019 frontier means no deep-net detour is warranted), and add the one belief mechanism with a principled monotonicity story—exact linear-Gaussian conjugacy—over its frozen latent (R4, R5), with a channel-agnostic linear-observation interface (R3). This section is written at design level; implementation details and all numbers are **[TODO:]**.

5.1 Frozen shallow-SOTA tower

The substrate is a certified shallow recommender—a RecVAE-class amortised encoder or a closed-form autoencoder [20, 21]—trained once on the full catalogue and then *frozen*. It exposes (a) item factor rows $\mathbf{Q}_i \in \mathbb{R}^d$, (b) a decoder read-out $s_i(\mathbf{u}) = \mathbf{Q}_i^\top \mathbf{u} + \beta_i$ with a learned bias β_i, and (c) the encoder that maps a full interaction set to a latent. Freezing is the wedge against Bıyık [2] (co-trained) and the guarantee that G0 (full strength preserved) is a property of the substrate, not of the belief layer. We deliberately do *not* extend the linear basis with concept columns (that stays in the item-indicator world and fails R3), nor learn the uncertainty head (no monotonicity story, and retraining risks G0). **[TODO:** Select and certify the frozen tower (paper-config RecVAE candidate at 0.3540/0.2497; report the certified full/tail NDCG@10 and the bridge number of §6.2).]

5.2 Analytic conjugate-Gaussian belief layer

Over the frozen latent we place a Gaussian belief on the user vector, $\mathbf{u} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, and model each answer as a noisy linear read-out along the queried direction:

$$y_j \;=\; \boldsymbol{\phi}_j^\top \mathbf{u} + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2). \tag{1}$$

Because the observation is linear-Gaussian, the posterior after $\mathcal{E}_t$ is Gaussian in closed form; maintaining it in *precision* form (a precision accumulator) makes the any-length fold-in a single running sum:

$$\boldsymbol{\Lambda}_t \;=\; \boldsymbol{\Lambda}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, \boldsymbol{\phi}_j \boldsymbol{\phi}_j^\top, \qquad \boldsymbol{\Lambda}_t\, \boldsymbol{\mu}_t \;=\; \boldsymbol{\Lambda}_0 \boldsymbol{\mu}_0 + \sum_{j=1}^{t} \sigma_j^{-2}\, y_j\, \boldsymbol{\phi}_j, \qquad \boldsymbol{\Sigma}_t = \boldsymbol{\Lambda}_t^{-1}. \tag{2}$$

Equation (2) is exactly the mechanism BCIE [24] uses for critiques over a frozen factorisation; our contribution is doing it over a *certified full-profile SOTA* tower with channel-agnostic tokens, and certifying the G0–G2 protocol on it. Three properties fall out by construction: it is a permutation-invariant *set* operation defined for every $t \geq 0$ (R2); each update adds a rank-1 term along $\boldsymbol{\phi}_{t+1}$, so $\boldsymbol{\Lambda}_t$ only grows and $\det \boldsymbol{\Sigma}_t$ only shrinks (G1); and conjugate updates are Blackwell/Good-1967 monotone in *expected utility*. The last is why R5 is plausible, but—because the tower's dot-product ranker does not act optimally under the belief—per-question NDCG monotonicity is *not* automatic and must be shown empirically (§7). **[TODO:** Fix $\boldsymbol{\Lambda}_0, \boldsymbol{\mu}_0$ (prior from the tower), the per-channel noise σ_j, and the confidence/temperature schedule; verify G0 holds at the belief mean on the full profile.]

5.3 Channel-agnostic tokens as linear observations

Every channel supplies a direction $\boldsymbol{\phi}_j \in \mathbb{R}^d$ and a value $y_j \in \mathbb{R}$ to Eq. (2); the update is identical across channels (R3):

- **Items.** $\boldsymbol{\phi}_j = \mathbf{Q}_i$ (the frozen item factor). A graded or signed answer sets y_j; a dislike is a negative y_j along the same direction—representable here precisely because the observation model (1) has a sign and a magnitude, unlike a one-hot indicator.
- **Out-of-catalogue concepts.** $\boldsymbol{\phi}_j$ is a concept direction in the latent (e.g. a centroid of member factors, or a text-adapter image of the concept), folded exactly like an item—no catalogue column required.
- **Entities / free text.** An embedding is projected into the latent by a fixed adapter and folded identically.
- **Continuous direction tokens.** A policy may emit any $\boldsymbol{\phi}_j \in \mathbb{R}^d$ directly (the continuous-action interface the companion chapter uses); the belief update does not distinguish it from a catalogue entity.

The signed/graded, out-of-catalogue, and continuous legs are the three channels inexpressible in the R1-bar basis (§4). [**TODO:** Specify the concept-direction and text-adapter constructions, the polarity/gradation encoding, and the acceptance tests (a like/dislike flip reverses the queried neighbourhood; a concept fold lifts its member items; monotone in t).]

6 Experimental design

6.1 Baseline bank

The baseline bank (Table 4) is three tiers: classic floors that every curve must clear (Dacrema et al. [7] demands them), the full-profile SOTA bar that decides G0, and the elicitation-compatible rivals that decide G1/G2. Reported anchors are on the systems' *own* splits and are not comparable to ours until bridged (§6.2); we print them only to show what "ties SOTA" will mean.

6.2 Evaluation protocol

The protocol items below instantiate the headline outcome gates (the certification bridge is C1; G0 and G2 as named); they are the cheap, always-run subset. The remaining per-build gates (G3–G9) and the load-bearing answer-model-transfer certification (C2), which close the outcome-gate false positives, are specified in full in the acceptance battery (Table 1) and its design-side document, and each headline result is reported against them.

(i) Certification bridge (C1). Our ML-25M ruler and the published bar are on *different* metrics and splits, and we do not conflate them. Concretely, our paper-config RecVAE realisation reaches full/tail NDCG@10 0.3540/0.2497 on the canonical ML-25M split—whereas the published EASE/RecVAE numbers are NDCG@100 on the ML-20M strong-generalisation split. These are not comparable, and no "ties SOTA" claim is made until the bridge is run. The bridge has two steps: (1) *reproduce* the published EASE and Mult-VAE/RecVAE numbers on the ML-20M Liang split (NDCG@100, Recall@20/50) in our own code, snapping to the reported values exactly [21, 13, 20]; (2) *port* the certified tower to the ML-25M harness and report full/tail NDCG@10 there. We keep ML-20M in the loop despite being older because the Liang 2018 split is a protocol lock-in the whole full-profile literature reports on; ML-25M is the newer corpus and, crucially, carries the genome tags the concept channel needs, so both datasets are required—one for comparability, one for the channels.

The first step is complete. On the ML-20M Liang strong-generalisation split our split statistics reproduce the published counts exactly (9,990,682 events / 116,677 train users / 20,108 items), and both anchor implementations **snap to their published values**: EASE at NDCG@100 0.4203 vs. the reported 0.420 [21], and RecVAE at 0.4425 vs. the reported 0.442, with Recall@20 (0.4144 vs. 0.414) and Recall@50 (0.5525 vs. 0.553) matching as well [20] (Mult-VAE/DAE certify the same way; [**TODO:** fill Mult-VAE/DAE snap deltas]). One closed-form and one neural reproduction to the third decimal certify that our split construction, metric implementation, and both implementation families are faithful to the published protocol—every ML-25M baseline number below therefore comes from certified code, and we spend no further space on it.

(ii) G0 — full-profile on ML-25M, items-only. On the baselines' native input (items only), evaluate full and tail NDCG@10 on the full ML-25M catalogue with a paired per-user bootstrap CI, confirming the instrument's belief mean reproduces the frozen tower and ties EASE/RecVAE on the shared harness. The item-only baselines are measured (Table 5).

A properly-trained RecVAE edges the closed-form frontier, and that is the bar. On our canonical ML-25M ruler our in-house paper-config RecVAE attains 0.3540 full / 0.2497 tail, the nominally strongest measured system—ahead of the closed-form leaders EASE (0.3476/0.2441) and EDLAE (0.3433/0.2395). The RecVAE-over-EASE margin is +0.0064 full, about 1.8 unpaired standard errors; we therefore state it as a *nominal* lead with a paired per-user bootstrap CI pending, not a settled win. All rows share the same split, cohort, tail definition, and NDCG@10 routine (verified by code inspection). The point of interest for the instrument is that on this ruler a properly-trained amortised VAE—not a closed-form model—sits at the top of the frontier; the instrument's R1 obligation is precisely to *tie this bar*, reproducing RecVAE-class full/tail accuracy through the frozen tower and then preserving it under the belief layer at G0. We report the whole frontier rather than excluding the stronger baselines.

(iii) G2 — per-question curves. Cold-start per-question NDCG@10 curves (full and tail) versus a random-question control. Items are foldable by all baselines; mixed channels (concepts, signed values, continuous tokens) are foldable only by our instrument, which is the point of the comparison—the rivals cap out at what an item-only interview can carry. Symmetric access is enforced where a baseline *can* fold a channel; where it structurally cannot, that is reported as an architecture limit, not hidden. [**TODO:** G2 curves for items-only (all methods) and mixed-channel (ours), full & tail, with CIs and the random control; G1 posterior-shrinkage diagnostic.]

(iv) Concept-as-pseudo-item ablation. To pre-empt "just add genre columns," we fold a concept into EASE/RecVAE as the centroid-of-members pseudo-item and compare against our concept channel. The expected finding is that a pseudo-item column is a lossy special case (it cannot carry sign, magnitude, or an out-of-catalogue direction), but we run it rather than assert it. [**TODO:** Ablation table: concept-as-pseudo-item fold into EASE/RecVAE vs. our concept channel, full & tail.]

Metrics. Full *and* tail NDCG@10 on ML-25M (the project reporting rule, scored with the decoder's learned bias, not a log-count popularity floor), plus NDCG@100 (and Recall@20/50) on the ML-20M bridge split for comparability with the published anchors. Every headline delta carries a paired per-user bootstrap CI on one shared harness. All result cells are [**TODO:**].

7 Limitations and threats to validity

- **Metric non-comparability until the bridge is run.** Our frontier number 0.3540/0.2497 is NDCG@10 on ML-25M; the published bar is NDCG@100 on ML-20M. Until §6.2 snaps our code to the published numbers (done for EASE/RecVAE, §6.2), *no* "ties SOTA" claim across the two rulers is licensed—the ML-25M and ML-20M numbers live on different rulers.
- **Monotonicity is empirical, not a theorem.** Conjugate-Gaussian updates are monotone in expected utility (Good 1967 / Blackwell) *only in expectation*, and only if the ranker acts optimally under the belief—which a fixed dot-product / NDCG ranker does not. R5 (and G2's rising curve) is therefore a *measured* property with a per-question CI, not a guarantee; a truthful answer can in principle still lower NDCG at a given step, and we report any such step rather than suppress it.
- **Simulator-based evaluation pending a human study.** G1/G2 are measured under a stated answer model (a simulator), which shares structure with the instrument's own latent; this risks circularity (the answer, the belief, and the metric are all functions of the same space). We add answer-noise ablations and a simulator-alignment limitation to every gate; the battery discharges this through certification C2 (the G2 gain must survive a held-out behavioural answer model plus injected noise, with a geometry-answer firewall) and treats the ≥ 1 human round-trip study (C3) as the load-bearing external validation—not yet run.
- **BERT4Rec replicability.** The sequence baseline's fold-in advantage is subject to a documented replicability caveat [18]; we tune it to the published protocol and flag any residual gap rather than treat a single run as authoritative.
- **Graph-filter admissibility.** GF-CF/BSPM/Turbo-CF [19, 4, 17] do not publish the ML-20M strong-generalisation NDCG; they enter Tier 2 only if re-run on our split, and are otherwise reported as an accuracy ceiling on other datasets, not as a bridged tie.
- **Absence of evidence.** The conjunction claim is "we are aware of none," bounded by our survey; a set-encoder / deep-set recommender that hits RecVAE-class full-profile accuracy *and* takes value-carrying mixed-type tokens would partly pre-empt R1$\wedge$R2$\wedge$R3. Our searches found no such reported system, which is absence of evidence, not proof of absence.

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Gate	One-line criterion	Failure class it excludes
Per-build gates (every build; all cheap)		
G0	Belief mean reproduces the frozen tower's full/tail NDCG@10 (tie or bit-identity).	Full-profile regression from the belief layer.
G1	$\mathbf{\Sigma}$ shrinks in expectation over confident answers (refusals exempt), directionally, calibrated to NLL.	Unusable / uncalibrated posterior.
G2	Cold full/tail curve beats random by AUC-at-budget and strictly rises; one true answer helps every channel.	Flat-model and out-of-envelope artifacts.
G3	Sign-flip lowers NDCG; value-neutralisation costs NDCG; quality monotone in intensity.	Sign-blind and intensity-blind folds.
G4	Precisions order refuse $<$ vague $<$ know-well; collapsing levels loses; a refusal barely moves belief.	Confidence-flat / mis-ordered ingestion.
G5	Concept lifts member items (AUC > 0.8) above its popularity projection; channel $\geq$ member-bag.	Non-specific / popularity-parasite concepts.
G6	Wrong-user, shuffled, placebo, and duplicated answers gain $\approx$ nothing.	Cardinality confound; existential leakage.
G8	Replacing $\mathbf{\Sigma}$ with $c\mathbf{I}$ degrades the deciding number; order- and re-ask-invariant.	Decorative $\mathbf{\Sigma}$; order-dependence.
G9	On a fixed user-independent question set the curve still rises from answer content.	Taste-leak: selection masquerading as ingestion.
One-time certifications (per instrument)		
C1	Implementations snap to published numbers on the canonical split (partially met).	Uncertified ruler / metric mismatch.
C2	G2 gain survives a held-out behavioural answer model + noise; geometry-answer firewall.	Circular answer model (self-simulator inversion).
C3	$\geq$ 1 study: rendered questions $\rightarrow$ graded human answers $\rightarrow$ NDCG (deferred).	Simulator-only external-validity gap.

Table 1: **The instrument acceptance battery.** Per-build gates G0–G9 (there is no standalone G7; the answer-model-transfer check is folded into C2) run on every build; certifications C1–C3 are one-time per instrument. The three headline outcome gates are G0/G1/G2; the semantic and artifact-control gates G3–G9 and the transfer certification C2 close the false positives that pass the outcome gates alone (text). The full specification, with each gate's testable form, is the design-side battery document.

Family	Representatives	R1	R2	R3	R4	R5	One-line verdict
(i) retrain / meta-learn per user	MeLU, TaNP [14], CDRNP	∼	✓	✗	∼	✗	R2/R4 precedent; not full-profile SOTA, not channel-open.
(ii) fold-in linear / closed-form	EASE [21], SLIM, GF-CF [19], BSPM [4], Turbo-CF [17]	✓	✓	✗	✗	∼	The full-profile bar (G0); item-indicator basis ⇒ R3 impossible.
(iii) amortised inference encoders	Mult-VAE [13], RecVAE [20], EDLAE [22], EDDI [16]	✓	✓	∼	∼	✗	The architecture we live in; VAEs item-only, EDDI right shape/wrong domain.
(iv) Bayesian belief over item space	Bıyık [2], ConTS [12], BCIE [24], ICF [25]	✗	✓	∼	✓	∼	The R4/R5 heart; belief layers, not full-profile towers.
(v) sequence models	SASRec [10], BERT4Rec [23]	∼	✓	✗	✗	✗	Interview ≠ ordered session; tokens item-only, order-sensitive.
(vi) LLM-as-recommender / CRS	PEBOL [1], GATE [11], LLM-CRS [9]	✗	✓	✓	∼	✗	Best R3 (open text); weakest R1 (weak collaborative signal).
CASPER (target)	**this chapter**	✓	✓	✓	✓	✓	**The conjunction; no published system occupies it.**

Table 2: Six-family taxonomy scored against R1–R5. ✓ met, ∼ partial, ✗ not met. Scores are our reading of each family's *published* claims and are qualitative; the last row is a target, not a result. R1 and {R3,R4,R5} are anti-correlated: families (ii)/(iii) hold the accuracy corner, family (iv) the belief corner, and none spans both. Adapted from the landscape analysis (c2).

(a) Classic / linear full-profile								
Most-Popular	✗	✓	✗	✗	∼	.1345 / .0226	measured	n/r (non-personalised floor)
Item-kNN (cosine)	✗	✓	✗	✗	∼	.2428 / .1297	measured	@100 ∼ 0.36 class [5]
PureSVD / iALS-WMF	✗	✓	✗	✗	∼	.2442 / .1853	measured (iALS)	@100 0.386 (WMF, ML-20M)
EASE [21]	✓	✓	✗	✗	∼	.3476 / .2441	measured	@100 0.420 (ours **PASS** +.0003)
EDLAE [22]	✓	✓	✗	✗	∼	.3433 / .2395	measured	@100 ≈ 0.42–0.44 class
GF-CF / BSPM / Turbo-CF [19, 4, 17]	∼	✓	✗	✗	∼	—	run planned (public code)	n/r (ML-20M not reported)
(b) Amortized / neural full-profile								
Mult-DAE [13]	✓	✓	✗	✗	✗	[TODO:]	queued	@100 0.419
Mult-VAE [13]	✓	✓	✗	∼	✗	[TODO:]	queued	@100 0.426
RecVAE [20] (ours, paper config)	✓	✓	✗	∼	✗	**.3540** / **.2497**	measured	@100 0.442 (ours **PASS** +.0005)
SASRec / BERT4Rec [10, 23]	∼	✓	✗	✗	✗	—	run planned (public code)	n/r (next-item; repl. [18])
TaNP [14] (MeLU / CDRNP line)	∼	✓	✗	✓	✗	—	best-effort planned	n/r (own cold-start benchmark)
(c) Elicitation / CRS systems — the belief corner								
EDDI / Partial-VAE [16]	✗	✓	∼	✓	✗	—	best-effort planned	n/r (non-recsys; UCI/MNIST)
Golbandi tree [8]	✗	✓	✗	✗	✗	.3065 / .1895	measured (core)	n/r (Netflix RMSE)
RBMF / Functional-MF [15, 26]	✗	✓	✗	✗	✗	—	best-effort planned	n/r (elicitation RMSE)
Bıyık soft-attributes [2]	✗	✓	∼	✓	∼	.2019 / .1200	measured (core)	n/r (own elicitation task)
ICF [25]	✗	✓	✗	✓	∼	—	core ≈ iALS (measured)	n/r (online CF; own protocol)
ConTS [12]	✗	✓	∼	✓	∼	.2019 / .1200	measured (core)	n/r (cold-start CRS metrics)
BCIE [24]	✗	✓	∼	✓	∼	—	not portable (needs their KG)	n/r (critique metrics)
UNICORN [6]	✗	✓	∼	✓	✗	—	not portable (needs their KG)	n/r (graph-RL CRS; own metrics)
PEBOL [1]	✗	✓	✓	∼	✗	—	not runnable at full catalogue	MRR@10 0.27 vs 0.17 (cold-start only)
GATE [11]	✗	✓	✓	✗	✗	—	not runnable (no ranker)	n/r (elicitation-only; no full ranking)
LLM zero-shot CRS [9]	✗	✓	✓	✗	✗	—	not runnable at scale	n/r (weak collaborative signal)
(d) Ours								
Set-encoder tower T2′ (ours)	∼	✓	∼	∼	∼	[TODO:]	training	n/a (in-house; graded-native tower)
Instrument (target)	✓	✓	✓	✓	✓	[TODO:]	pending	n/a (this chapter)

Table 3: **Master table: every reasonable system by R1–R5 and by measured full-profile accuracy on one shared ruler.** ✓ met, ∼ partial, ✗ not met (our reading of each system's *published* claims, consistent with Table 2). *full/tail NDCG@10* is on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; tail = top-33%-train-mass head mask, 190 head items), scored with the decoder's learned bias. The Bıyık and ConTS rows share one measured posterior core (belief-MF, best-effort; substitutions per the no-dash paragraph), so both print 0.2019/0.1200; the Golbandi row prints its node-recommender core. Statuses are *measured* (run on this ruler), *queued* ([TODO:], run pending), *run planned / best-effort planned* (replication on this ruler in the next experimental round; best-effort = with stated substitutions), *superseded* (a measured successor row subsumes it), or *not runnable / not portable* (impossible by construction; mechanism in the text). The final column is each system's own published full-profile anchor on its own split/metric—*not* comparable to our ruler until bridged (§6.2); "n/r" where the literature reports none. The RecVAE row is our own paper-config training run on this

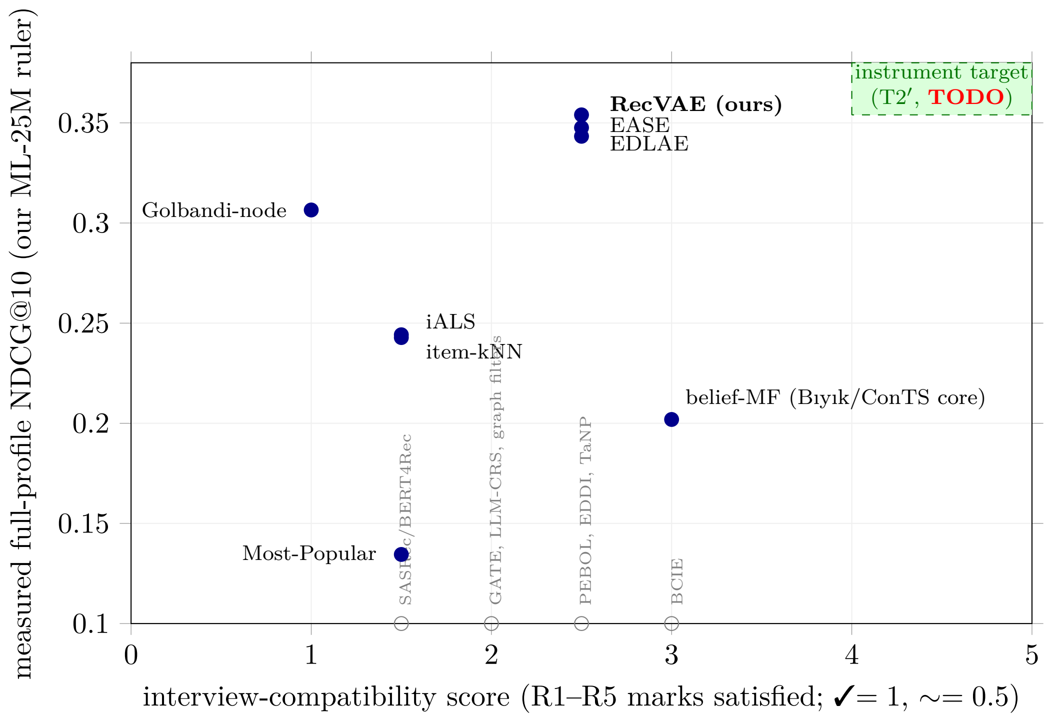

Figure 1: **The gap map (measured axes).** y-axis: measured full-profile NDCG@10 on our canonical ML-25M ruler. x-axis: an *interview-compatibility score*, defined as the number of R1–R5 marks a system carries in the master table (Table 3), scoring ✓=1 and ∼=0.5. This score is a *coarse ordinal capability count, not a measurement*, so the x-axis is qualitative even though the y-axis is measured. Filled markers are systems with a full-profile number on this ruler (belief-MF is the shared Bıyık/ConTS posterior core; Golbandi-node is the tree's best-effort node recommender). Hollow markers, pinned at the axis floor, are systems for which *no full-profile number exists*—unrun, unrunnable at full catalogue, or not portable (see the master-table dispositions); their height is not meaningful, only their x. The shaded top-right box is the instrument target ($x \geq 4$ and full-profile at the frontier bar), still **[TODO:]**. The measured points descend from left to right: accuracy peaks at the item-only accuracy corner (x=2.5: RecVAE/EASE/EDLAE) and falls as interview-compatibility rises, with the highest-compatibility measured core (belief-MF at x=3.0) sitting near the bottom of the accuracy column. The nearest published rival to the target is Bıyık et al. [2] (highest compatibility of any system with an actionable belief), whose runnable core nonetheless measures at 0.2019 full—far below the frontier bar.

Baseline	Type	Reported anchor	Why the slot
Tier 1 — classic floors (must clear)			
Most-Popular	non-personalised	popb floor (our harness)	the "did we learn anything" line; G2 must beat it cold.
Item-kNN (cosine)	neighbourhood	ML-20M NDCG@100 ~ 0.36 class	instant fold-in; Cremonesi-era strong-simple [5].
PureSVD / iALS-WMF	linear MF fold-in	WMF ML-20M NDCG@100 0.386	cheap fold-in reference.
SLIM	sparse linear item-item	ML-20M NDCG@100 0.401	direct EASE ancestor.
Tier 2 — full-profile SOTA (the R1 / G0 bar)			
EASE [21]	closed-form item-item	NDCG@100 0.420	the bar to tie; item-only shows the R3 gap.
RecVAE [20]	amortised VAE	NDCG@100 0.442	top of the ML-20M split; our in-house frontier.
Mult-VAE / DAE [13]	VAE / DAE	NDCG@100 0.426 / 0.419	canonical amortised anchor.
EDLAE [22]	denoising higher-order EASE	$\approx$ RecVAE class	confirms the flat frontier.
GF-CF / BSPM / Turbo-CF [19, 4, 17]	closed-form graph filter	SOTA on Gowalla/Yelp (ML-20M not reported)	admit only if re-run on our split.
SASRec / BERT4Rec [10, 23]	self-attentive sequence	strong next-item (not this split)	sequence fold-in; watch replicability [18].
Tier 3 — elicitation-compatible rivals (decide G1/G2)			
EDDI / Partial-VAE [16]	set-encoder + info-gain	non-recsys (UCI/MNIST)	the R2 architecture; myopic info-gain baseline.
Bıyık soft-attributes [2]	Gaussian belief + EVOI	own elicitation task	the central threat; frozen-vs-co-trained wedge.
ConTS [12]	Thompson bandit, item+attr arms	cold-start CRS metrics	unified-space + uncertainty; fixed schema.
BCIE [24]	conjugate-Gaussian critique fold	critique metrics	exact-conjugacy-over-frozen precedent.
PEBOL [1]	Bayesian-opt NL-PE (LLM)	MRR@10 0.27 vs 0.17	load-bearing PE baseline; per-item Beta.
Golbandi tree [8]	RMSE decision-tree interview	Netflix RMSE	the static-tree interview comparator.
RBMF / Functional-MF [15, 26]	MF-embedding seeds	elicitation RMSE	interview-seed line; cite-and-differentiate.

Table 4: Baseline bank. Anchors are on each system's own split/metric and are *not* comparable to ours until the bridge (§6.2) is run; G0 is decided against EASE+RecVAE, G1/G2 against the Tier-3 rivals plus a random-question control. Any deep-net "SOTA" not snapping to Tier-2 numbers on our split is inadmissible [7].

Method	full NDCG@10	tail NDCG@10	note
Most-Popular	0.1345	0.0226	@100 0.1961; non-personalised floor
belief-MF (Bıyık/ConTS core)	0.2019	0.1200	@100 0.2886; best-effort posterior core
Item-kNN (cosine)	0.2428	0.1297	@100 0.3200; neighbourhood fold-in
iALS-WMF	0.2442	0.1853	@100 0.3704; MF fold-in
Golbandi node recommender	0.3065	0.1895	@100 0.3958; best-effort tree core
EDLAE [22]	0.3433	0.2395	@100 0.4330; val p=0.4
EASE [21]	0.3476	0.2441	@100 0.4375
RecVAE (ours, paper config) [20]	**0.3540**	**0.2497**	@100 0.4537; nominal frontier
Ours			
Set-encoder tower T2′ (ours)	**[TODO:]**		graded-native interview tower (training)

Table 5: G0 — full-profile, items-only, on the canonical Liang-recipe strong-generalization ML-25M split (catalog restricted to the 18,430-item project catalog; train 140,768 / val 10k / test 10k users; per-user 80/20 fold-in; seed 98765), NDCG@10 (full and tail); tail = top-33%-train-mass head mask (190 head items); scored with the decoder's learned bias. The belief-MF and Golbandi rows are best-effort cores of the belief-corner systems (§3). All rows reuse the canonical harness verbatim, so they are on one ruler.